PROGRAM-4

1. Salary Components

Earnings:

- Basic Pay
- Allowances (HRA, DA, TA, etc.)
- Overtime Pay
- Incentives/Bonus

Gross Salary Formula

$$\text{Gross Salary} = \text{Basic Pay} + \text{Allowances} + \text{Overtime Pay} + \text{Incentives}$$

2. Progressive Deductions

Example deduction policy:

Gross Salary Range	Tax Deduction
Up to ₹25,000	5%
₹25,001 - ₹50,000	10%
Above ₹50,000	15%

Additional deductions:

- Provident Fund (PF): 12% of Basic Pay
- Professional Tax: ₹200

3. Net Salary Calculation

- $\text{Total Deductions} = \text{Tax} + \text{PF} + \text{Professional Tax}$
  $\text{Net Salary} = \text{Gross Salary} - \text{Total Deductions}$

4. Algorithm

- Input employee details and salary components.
- Calculate gross salary.
- Determine applicable tax slab.
- Calculate tax, PF, and professional tax.
- Compute total deductions.
- Calculate net salary.
- Generate formatted salary statement.
- Display all earnings, deductions, and net pay.

5. Sample Calculation

Input

- Basic Pay = ₹40,000
- Allowances = ₹8,000

- Overtime Pay = ₹2,000
- Incentives = ₹5,000

Gross Salary

$$40{,}000 + 8{,}000 + 2{,}000 + 5{,}000 = ₹55{,}000$$

Deductions

- Tax (15%) = ₹8,250
- PF (12% of ₹40,000) = ₹4,800
- Professional Tax = ₹200
- **Total Deductions**
- $8{,}250 + 4{,}800 + 200 = ₹13{,}250$

- **Net Salary**
- $55{,}000 - 13{,}250 = ₹41{,}750$

- **Formatted Salary Statement**
- ```
  ---------------------------------
          SALARY STATEMENT
  ---------------------------------
  Basic Pay           : ₹40,000
  Allowances          : ₹ 8,000
  Overtime Pay        : ₹ 2,000
  Incentives          : ₹ 5,000
  ---------------------------------
  Gross Salary        : ₹55,000
  ---------------------------------
  Tax Deduction       : ₹ 8,250
  PF Deduction        : ₹ 4,800
  Professional Tax    : ₹   200
  ---------------------------------
  Total Deductions    : ₹13,250
  ---------------------------------
  Net Salary          : ₹41,750
  ---------------------------------
  ```

programm

```
# Employee Payroll and Deduction Analysis

# Input employee details
name = input("Enter employee name: ")
employee_id = input("Enter employee ID: ")

basic_pay = float(input("Enter basic pay: "))
allowances = float(input("Enter allowances: "))
overtime = float(input("Enter overtime amount: "))
incentives = float(input("Enter incentives: "))

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# Calculate gross salary
gross_salary = basic_pay + allowances + overtime + incentives

# Calculate progressive deductions
if gross_salary <= 20000:
    deduction = gross_salary * 0.05

elif gross_salary <= 40000:
    deduction = (20000 * 0.05) + ((gross_salary - 20000) * 0.10)

else:
    deduction = (20000 * 0.05) + (20000 * 0.10) + ((gross_salary - 40000) * 0.15)

# Calculate net salary
net_salary = gross_salary - deduction

# Display salary statement
print("\n========== SALARY STATEMENT ==========")
print("Employee Name   :", name)
print("Employee ID     :", employee_id)
print("-------------------------------------")
print("Basic Pay       : Rs.", format(basic_pay, ".2f"))
print("Allowances      : Rs.", format(allowances, ".2f"))
print("Overtime        : Rs.", format(overtime, ".2f"))
print("Incentives      : Rs.", format(incentives, ".2f"))
print("-------------------------------------")
print("Gross Salary    : Rs.", format(gross_salary, ".2f"))
print("Total Deduction : Rs.", format(deduction, ".2f"))
print("Net Salary      : Rs.", format(net_salary, ".2f"))
print("=====================================")
```

OUTPUT:

Enter employee name: hanifa

Enter employee ID: 26001021

Enter basic pay: 500

Enter allowances: 622

Enter overtime amount: 455

Enter incentives: 789

========== SALARY STATEMENT ==========

Employee Name : hanifa

Employee ID : 26001021

Basic Pay : Rs. 500.00

Allowances : Rs. 622.00

Overtime : Rs. 455.00

Incentives : Rs. 789.00

Gross Salary : Rs. 2366.00

Total Deduction : Rs. 118.30

Net Salary : Rs. 2247.70

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